ATOM Instruments

UNIVERSAL TEMPERATURE CALIBRATOR

ATOM 14+

The ATOM Universal Temperature Source & Measure Calibrator 14+ is a versatile multifunction calibrator designed for precise measurement and simulation of thermocouple, RTD, voltage, resistance, and making it ideal for calibration, maintenance, and industrial process control applications.

NABL CALIBRATION CERTIFICATE INCLUDED

KEY FEATURES

- Universal source & measurement functions
- Supports Thermocouple and RTD calibration
- Measures & sources Voltage, Resistance
- High-accuracy calibration performance
- Dual LCD display with backlight
- Compact, portable and user-friendly design
- Basic Accuracy ±0.02%

HIGH ACCURACY | RELIABLE PERFORMANCE | INDUSTRIAL GRADE | PORTABLE DESIGN

ISO CERTIFIED | 1 YEAR WARRANTY

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MEASURE MODE

- Accuracy is expressed as ± (percentage of reading + percentage of range).

Function	Test Current	Range	Resolution	Accuracy
OHM	Approximately 1 mA	0.00 Ω ~ 550.00 Ω	0.01 Ω	±(0.05 + 0.02)
	Approximately 0.1 mA	0.0000 kΩ ~ 5.000 kΩ	0.1 Ω	±(0.05 + 0.02)

Function	Reference	Range	Resolution	Accuracy
DC Voltage	50 mV	-5.000 ~ 55.000 mV	1 μV	±(0.02 + 0.02)
	500 mV	-50.00 ~ 550.00 mV	10 μV	±(0.02 + 0.01)

Function	Type	Range	Resolution	Accuracy
TC	R	0°C ~ 1767°C	1°C	0–500°C: ±1.8°C; 500–1767°C: ±1.5°C
	S	0°C ~ 1767°C	1°C	0–500°C: ±1.8°C; 500–1767°C: ±1.5°C
	K	-100.0°C ~ 1372.0°C	0.1°C	-100–0°C: ±1.2°C; 0–1372°C: ±0.8°C
	E	-50.0°C ~ 850.0°C	0.1°C	-50–0°C: ±0.9°C; 0–850°C: ±1.5°C
	J	-60.0°C ~ 1120.0°C	0.1°C	-60–0°C: ±1.0°C; 0–1120°C: ±0.7°C
	T	-100.0°C ~ 400.0°C	0.1°C	-100–0°C: ±1.0°C; 0–400°C: ±0.7°C
	N	-200.0°C ~ 1300.0°C	0.1°C	-200–0°C: ±1.5°C; 0–1300°C: ±0.9°C

Function	Reference	Range	Resolution	Accuracy
TC	B	600°C ~ 1820°C	1°C	600–800°C: ±2.2°C; 800–1000°C ±1.8°C; 1000–1820°C: ±1.4°C
	L	-60°C ~ 900°C	0.1°C	-60–0°C: ±0.7°C; 0–900°C: ±0.5°C
	U	-100°C ~ 600°C	0.1°C	-100–0°C: ±0.7°C; 0–600°C: ±0.5°C

Function	Reference	Range	Resolution	Accuracy
RTD	Pt100-385	-200.0°C ~ 800.0°C	0.1°C	-200–0°C: ±0.5°C; 0–400°C: ±0.7°C; 400–800°C: ±0.8°C
	Pt200-385	-200.0°C ~ 630°C	0.1°C	-200–100°C: ±0.8°C 100–300°C: ±0.3°C 300–630°C: ±1.0°C
	Pt500-385	-200.0°C ~ 630°C	0.1°C	
	Pt1000-385	-200°C ~ 630°C	0.1°C	-200-100: ±0.8°C 100-300°C: ±0.9°C 300–630°C: ±1.0°C
	Cu10	-100°C ~ 260°C	0.1°C	-100–260°C: ±1.8°C
	Cu50	-50°C ~ 150°C	0.1°C	-50 to 150°C: ±0.7°C

Function	Reference	Range	Resolution	Remark
CONTINUITY	500Ω, Approximately 1 mA Test Current	≤ 50Ω (Sound)	0.01Ω	Excitation Current Approximately: 1mA

SOURCE MODE

- Accuracy is expressed as ± (percentage of reading + percentage of range).

Function	Reference	Range	Resolution	Accuracy
DC Voltage	100 mV	-10.000 mV ~ 110.000 mV	1 μV	±(0.02 + 0.01)
	1000 mV	-100.00 mV ~ 1100.00 mV	10 μV	±(0.02 + 0.01)

Function	Reference	Range	Resolution	Accuracy
Resistance	400 Ω	0.00 Ω ~ 400.00 Ω	0.01 Ω	±(0.02 + 0.025)
	4 kΩ	0.0000 kΩ ~ 4.0000 kΩ	0.1 Ω	±(0.05 + 0.025)

Function	Type	Range	Resolution	Accuracy
TC	R	0°C ~ 1767°C	1°C	0–100°C: ±1.5°C; 100–1767°C: ±1.2°C
	S	0°C ~ 1767°C	1°C	0–100°C: ±1.5°C; 100–1767°C: ±1.2°C
	K	-200.0°C ~ 1372.0°C	0.1°C	-200–100°C: ±0.6°C; -100–400°C: ±0.5°C; 400–1200°C: ±0.7°C; 1200–1372°C: ±0.9°C
	E	-200.0°C ~ 1000.0°C	0.1°C	-200–100°C: ±0.6°C; -100–600°C: ±0.5°C; 600–1000°C: ±0.4°C
	J	-200.0°C ~ 1200.0°C	0.1°C	-200–100°C: ±0.6°C; -100–800°C: ±0.5°C; 800–1200°C: ±0.7°C
	T	-250.0°C ~ 400.0°C	0.1°C	-250–400°C: ±0.6°C
	N	-200.0°C ~ 1300.0°C	0.1°C	-200–100°C: ±1.0°C; -100–900°C: ±0.7°C; 900–1300°C: ±0.8°C
	B	600°C ~ 1820°C	0.1°C	600–800°C: ±1.5°C; 800–1820°C: ±1.1°C
	L	-200.0°C ~ 900.0°C	0.1°C	-200–0°C: ±0.7°C; 0–900°C: ±0.5°C
	U	-200.0°C ~ 600.0°C	0.1°C	-200–0°C: ±0.7°C; 0–600°C: ±0.5°C

Function	Type	Range	Resolution	Accuracy
RTD	Pt100-385	-200.0°C ~ 800.0°C	0.1°C	-200–0°C: ±0.3°C; 0–400°C: ±0.5°C; 400–850°C: ±0.8°C
	Pt200-385	-200.0°C ~ 630.0°C	0.1°C	-200–100°C: ±0.8°C; 100–300°C: ±0.9°C; 300–630°C: ±1.0°C
	Pt500-385	-200.0°C ~ 630.0°C	0.1°C	-200–100°C: ±0.4°C; 100–300°C: ±0.5°C; 300–630°C: ±0.7°C
	Pt1000-385	-200.0°C ~ 630.0°C	0.1°C	-200–100°C: ±0.2°C; 100–300°C: ±0.5°C; 300–630°C: ±0.7°C
	Cu10	-100.0°C ~ 250.0°C	0.1°C	-100–250°C: ±1.8°C
	Cu50	-50.0°C ~ 150.0°C	0.1°C	-50–150°C: ±0.6°C